Technical Report

CMPG 325 Milestone 3

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Field	Detail
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Student number	52827097
Module	CMPG 325 Computer Networks
Project ID	CMPG325-2026-121
Client	Platinum Sound Recording Studio, Rustenburg (CLI-121)
Industry	Entertainment
Assigned feature	SSH secure device management
Change request	CR-14: after-hours contractor wireless access
Packet Tracer file	52827097_SEDZE_CMPG325-2026-121.pkt
GitHub repository	https://github.com/wantedProgrammer/Computer-Networks

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1. Executive summary

Platinum Sound Recording Studio now has a working, segmented and secured network built in Cisco Packet Tracer and verified by 15 tests which all passed. The design separates the studio's three work areas, its servers and its contractors into five VLANs on the assigned 192.168.52.0/24 block.

The delivered network provides:

- **Separation by function -** Admin, Recording Studios, Production, Servers and Guest each have their own VLAN and subnet, sized with VLSM.
- **Fast internal routing -** A Layer 3 core switch routes between VLANs, so Production pulls session files from the server without passing through the edge router.
- **Automatic addressing -** DHCP on the core switch serves all user VLANs. Servers keep fixed addresses.
- **Internet access -** The edge router translates all internal traffic to one public address using NAT overload.
- **CR-14 contractor Wi-Fi -** A WPA2 network in its own VLAN gives cleaning and security contractors internet only, an ACL blocks every internal VLAN.
- **Secure management -** All five infrastructure devices accept only encrypted SSHv2 sessions, and only from the Admin/Management VLAN. Telnet is disabled.

One requirement could only be partly simulated: Packet Tracer does not support time-based ACLs, so the "after-hours" window for CR-14 is documented as real Cisco IOS configuration.

2. Client requirements analysis

The studio needs staff areas kept apart shared access to session files, secure remote management and limited Wi-Fi for after-hours contractors. Platinum Sound is a boutique recording studio in Rustenburg. It handles client-owned audio so protecting session files is a business requirement, a technical one.

2.1 Functional requirements

#	Requirement	How the design meets it
F1	Wired connectivity for Admin, Recording Studios and Production, kept logically separate	VLANs 10, 20 and 30, one access switch per area
F2	Central storage for session files, reachable from Admin and Production	File server in VLAN 40, routed by the core switch
F3	Internet access for staff	Edge router with NAT overload
F4	Secure, auditable device management with no cleartext protocols	SSHv2 only, local accounts, VTY limited to VLAN 10
F5	Limited wireless access for after-hours contractors (CR-14)	WPA2 guest SSID in VLAN 50, internet-only ACL

2.2 Non-functional requirements

Requirement	Target	How it is met
Scalability	Room to grow without re-addressing	124 addresses (192.168.52.132–.255) left free; each /27 holds 30 hosts
Security	Contractors and staff areas cannot reach what they do not need	Guest ACL, SSH access-class, shut unused ports
Performance	Fast access to large audio files between VLANs	Inter-VLAN routing on the core switch backplane
Availability	One area's failure does not take down the others	Separate access switch per department
Maintainability	Easy to understand and troubleshoot	One VLAN per function, consistent naming, configs in GitHub

2.3 Change request CR-14

The client's change request reads: "After-hours cleaning and security contractors need limited wireless access." Three conditions follow from it. The access must be wireless, it must be limited to the internet not studio systems, and it should apply after hours. The first two are fully implemented, the third is documented in section 7.

3. Assumptions and scope

The design rests on six assumptions made where the brief was silent. Each keeps the scope realistic for a small studio.

Assumption	Reason
Three work zones: Admin, Recording Studios (A and B in one VLAN), Production/Editing	Typical layout for a small-to-medium studio. Business admin sits apart from creative spaces
15 - 25 devices per zone, so a /27 (30 hosts) per zone	A boutique studio, not an enterprise. VLSM sized to real scale with room to grow
One shared file server for session files	Studios centralize project files for backup and cross-studio access
Contractors bring their own phones or tablets and need internet only	Matches "limited wireless access" in CR-14
No VoIP or phone system	None was requested, so none was designed
VLAN 10 doubles as the management VLAN	Admin staff manage the network. A separate management VLAN adds cost without a stated need

Out of scope: the ISP router and internet server in the Packet Tracer file are a test harness, added only to prove internet access and NAT. They use RFC 5737 documentation addresses 203.0.113.0/30 and 198.51.100.0/24 and are not part of the client's network.

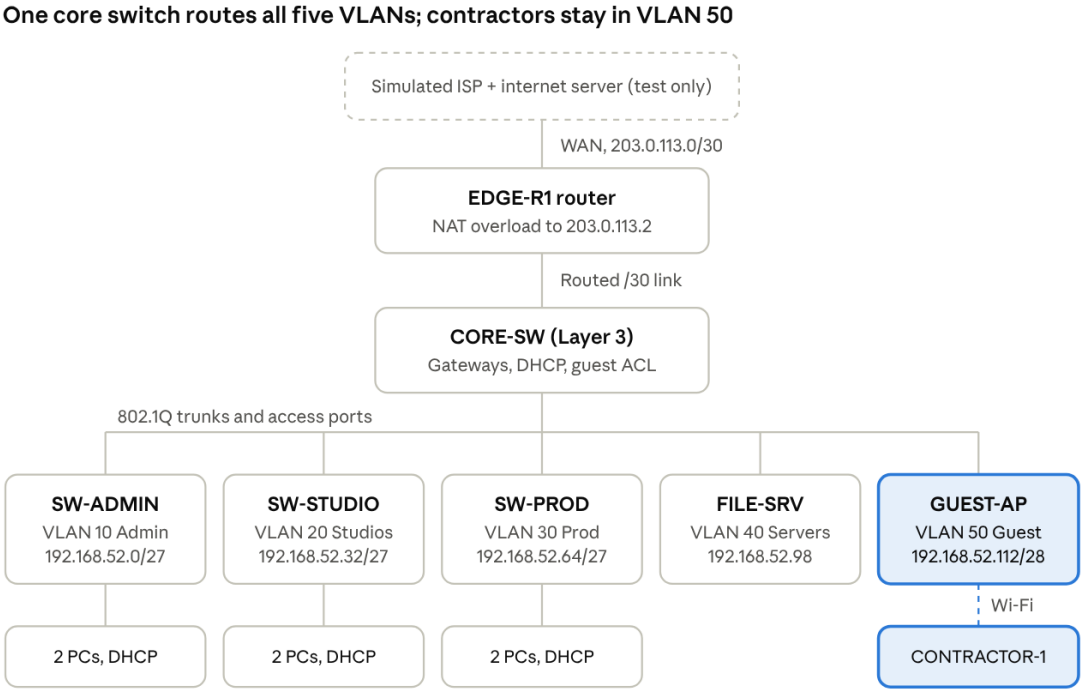
4. Network design

The network is a collapsed-core design: one Layer 3 core switch routes between all five VLANs, three access switches serve the staff areas, and one edge router connects to the internet.

4.1 Design decisions

Decision	Chosen	Alternative rejected	Why
Inter-VLAN routing	Layer 3 core switch (SVIs)	Router-on-a-stick	Routing in the switch backplane is faster than sending every inter-VLAN packet up one router link and back, which matters for large audio files
Router role	Edge only: WAN and NAT	Router also routes internal VLANs	Keeps the internet-facing device simple and its attack surface small
Access layer	One switch per department	One large shared switch	A failed switch takes down one area, not the whole studio
Trunks	Only the department VLAN + VLAN 10 per trunk	All VLANs on every trunk	A switch cannot carry traffic for VLANs it does not serve
Contractor Wi-Fi	Separate AP, own VLAN, ACL at its gateway	Guest SSID on a shared AP	No staff network is ever broadcast to outsiders
Management	SSH from VLAN 10 only	Telnet, or SSH from anywhere	Encrypted, and only reachable from admin machines

4.2 Topology



logical topology - 5 VLANs, 1 core switch

Every inter-VLAN packet turns around at CORE-SW, only internet traffic continues to EDGE-R1. Contractors join through their own access point and VLAN, where the guest ACL keeps them off the four internal VLANs.

5. IP addressing plan

The assigned 192.168.52.0/24 block is split with VLSM into five VLAN subnets and one point-to-point link, using 132 addresses and leaving 124 free for growth.

VLAN	Purpose	Subnet	Mask	Usable range	Gateway	Broadcast	Assignment
10	Admin / Management	192.168.52.0/27	255.255.255.224	.1–.30	.1	.31	DHCP from .10; switch SVIs .2–.4
20	Recording Studios	192.168.52.32/27	255.255.255.224	.33–.62	.33	.63	DHCP from .38
30	Production / Editing	192.168.52.64/27	255.255.255.224	.65–.94	.65	.95	DHCP from .70
40	Servers	192.168.52.96/28	255.255.255.240	.97–.110	.97	.111	Static; FILE-SRV .98
50	Guest wireless (CR-14)	192.168.52.112/28	255.255.255.240	.113–.126	.113	.127	DHCP from .116
–	Router ↔ core link	192.168.52.128/30	255.255.255.252	.129–.130	–	.131	EDGE-R1 .129, CORE-SW .130
–	Reserved for growth	192.168.52.132–.255	–	–	–	–	Unallocated

Why the subnets are sized this way:

- **/27 for staff VLANs -** Each zone expects 15 - 25 devices. A /27 gives 30 usable hosts, a /28 (14 hosts) would be too small and a /26 (62) would waste half the space.
- **/28 for servers and guests -** Few servers are needed, and only a handful of contractors are on site at night so 14 hosts each is enough.
- **/30 for the router link -** A point-to-point link needs exactly two addresses, so the smallest usable subnet avoids waste.
- **Largest subnets first -** Allocating the /27s, then /28s, then the /30 keeps every subnet on a clean boundary and leaves one unbroken free block at the top.
- **One summary route -** VLANs 10 - 50 all fall inside 192.168.52.0/25, so the edge router needs only one static route back to the inside network.

6. Implementation

The network was built in seven phases each tested before the next so any fault could be traced to the last change.

1. Place and cable devices
2. Access switches: VLANs and trunks
3. Core switch: SVIs, inter-VLAN routing, DHCP
4. Edge router, NAT and simulated ISP
5. File server and end devices
6. CR-14 contractor wireless and guest ACL
7. SSH on all infrastructure devices

6.1 Devices

Device	Packet Tracer model	Role	Management IP
EDGE-R1	Cisco 2911	Edge gateway, NAT	192.168.52.129
CORE-SW	Cisco 3560-24PS	Inter-VLAN routing, DHCP, guest ACL	192.168.52.1
SW-ADMIN	Cisco 2960-24TT	Access switch, VLAN 10	192.168.52.2
SW-STUDIO	Cisco 2960-24TT	Access switch, VLAN 20	192.168.52.3
SW-PROD	Cisco 2960-24TT	Access switch, VLAN 30	192.168.52.4
FILE-SRV	Server-PT	Session file storage (FTP)	192.168.52.98
GUEST-AP	AccessPoint-PT	Contractor Wi-Fi	–
6 × PC	PC-PT	Two staff PCs per zone	DHCP
CONTRACTOR-1	Smartphone-PT	Contractor device	DHCP

6.2 Access layer

Each access switch puts ports Fa0/1 - 20 in its department VLAN and trunks only that VLAN plus VLAN 10 to the core. Unused ports Fa0/21 - 24 are shut down so nobody can plug in an unknown device. PortFast lets PCs connect immediately instead of waiting about 30 seconds for spanning tree.

```
interface range fa0/1 - 20
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
interface range fa0/21 - 24
 shutdown
interface g0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,30
```

6.3 Core switch: inter-VLAN routing and DHCP

CORE-SW holds one SVI per VLAN as that VLAN's gateway, with `ip routing` enabled. Its uplink to the router is a routed port (`no switchport`) on the /30 link, and a default route points to EDGE-R1.

DHCP pools on the core switch serve VLANs 10, 20, 30 and 50. Each pool excludes its gateway and reserved addresses, so a PC can never be handed a switch's management IP. Servers in VLAN 40 use static addresses because clients must always find them in the same place.

```
ip routing
interface g0/1
 no switchport
 ip address 192.168.52.130 255.255.255.252
interface vlan 20
 ip address 192.168.52.33 255.255.255.224
ip route 0.0.0.0 0.0.0.0 192.168.52.129
ip dhcp excluded-address 192.168.52.33 192.168.52.37
ip dhcp pool STUDIO
 network 192.168.52.32 255.255.255.224
 default-router 192.168.52.33
 dns-server 198.51.100.10
```

6.4 Edge router and NAT

EDGE-R1 has one summary route to the inside 192.168.52.0/25 and a default route to the ISP. NAT overload PAT translates every internal address to the router's single outside address, 203.0.113.2 telling sessions apart by port number.

```
interface g0/0
 ip address 192.168.52.129 255.255.255.252
 ip nat inside
interface g0/1
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
ip route 192.168.52.0 255.255.255.128 192.168.52.130
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

```
access-list 1 permit 192.168.52.0 0.0.0.127
ip nat inside source list 1 interface g0/1 overload
```

6.5 File server

FILE-SRV sits in VLAN 40 at 192.168.52.98/28 with FTP enabled. Production and Admin reach it through the core switch in one routed hop. Full running configurations for all five infrastructure devices are in the repository's `/configs` folder.

7. CR-14: contractor wireless access

Contractors get internet-only Wi-Fi in their own VLAN, an inbound ACL on the VLAN 50 gateway drops everything aimed at the studio's internal network. Before the ACL, a contractor phone could ping the file server. Afterwards the same ping failed while internet access kept working.

7.1 Design

Element	Setting	Why
VLAN	50, 192.168.52.112/28	Keeps contractors in their own broadcast domain
Access point	GUEST-AP on CORE-SW Fa0/20	A separate AP means no staff SSID is shared with outsiders
SSID	PSR-Contractors	Clear name for contractors to find
Security	WPA2-PSK with AES	Encrypts the air link; WEP and open Wi-Fi are not acceptable
Addressing	DHCP pool GUEST, from .116	Contractors' own devices need no setup
Policy	Extended ACL GUEST-IN, inbound on interface Vlan50	Filters traffic as it enters the network, before it can be routed

7.2 The GUEST-IN access list

```
ip access-list extended GUEST-IN
 remark CR-14 contractor wireless - internet only
 permit udp any eq bootpc any eq bootps
 deny ip 192.168.52.112 0.0.0.15 192.168.52.0 0.0.0.255
 permit udp 192.168.52.112 0.0.0.15 any eq domain
 permit tcp 192.168.52.112 0.0.0.15 any eq www
 permit tcp 192.168.52.112 0.0.0.15 any eq 443
 permit icmp 192.168.52.112 0.0.0.15 any echo
 deny ip any any
interface vlan 50
 ip access-group GUEST-IN in
```

The order of the entries is the design, because an ACL stops at its first match:

1. **DHCP first -** A new device has no IP yet, so its request must be allowed before anything else.
2. **Then deny all internal traffic -** Everything to 192.168.52.0/24 is dropped. All four staff and server VLANs, the router link and the gateway itself.
3. **Then allow only DNS, web and ping -** Because the internal deny comes first, these can only ever reach outside addresses.
4. **Explicit final deny -** IOS denies everything else anyway. Writing it out gives a hit counter for blocked traffic.

7.3 Results

After testing, `show access-lists` recorded 8 matches on the internal deny two blocked pings of 4 packets, 1 DNS lookup, 5 web packets and 4 internet pings. The DHCP line showed 0 matches only because the contractor's device had leased its address before the ACL was applied.

7.4 The "after-hours" limit

Packet Tracer does not support the `time-range` command, so the after-hours window cannot be simulated. On real Cisco IOS it would be added like this:

```
time-range CONTRACTOR-HOURS
 periodic daily 18:00 to 23:59
 periodic daily 0:00 to 6:00
!
ip access-list extended GUEST-IN
 permit udp 192.168.52.112 0.0.0.15 any eq domain time-range CONTRACTOR-HOURS
 permit tcp 192.168.52.112 0.0.0.15 any eq www time-range CONTRACTOR-HOURS
 permit tcp 192.168.52.112 0.0.0.15 any eq 443 time-range CONTRACTOR-HOURS
```

The window is split into two `periodic` entries because one entry cannot wrap past midnight. Outside these hours the permit lines stop matching, so contractor traffic falls through to the final deny.

8. Assigned feature: SSH secure management

All five infrastructure devices run SSHv2 with Telnet disabled and only hosts in VLAN 10 can open a session. In testing, Admin logged into three devices over an AES-128 encrypted session, while a Studio PC was refused before it was even asked for a password.

8.1 Why SSH, and why these controls

Telnet sends usernames, passwords and every command in plain text so anyone capturing traffic on the LAN could read the admin password. SSH encrypts the whole session. Encryption alone is not enough, though. The design also limits who can try to log in and how many attempts they get. The controls work in layers:

Layer	Control	Command
Who can connect	Only VLAN 10 (192.168.52.0/27) may reach the management lines	access-list 10 permit 192.168.52.0 0.0.0.31 + access-class 10 in
How they connect	SSH only; Telnet refused	transport input ssh
Protocol strength	SSHv1 disabled; 1024-bit RSA keys	ip ssh version 2, crypto key generate rsa … modulus 1024
Who they are	Local username with a hashed password	username netadmin privilege 15 secret …, login local
Brute-force protection	60-second login window, 3 attempts	ip ssh time-out 60, ip ssh authentication-retries 3
Idle sessions	Closed after 5 minutes	exec-timeout 5 0
Physical access	Console also needs a login	line console 0 + login local
Legal notice	Warning shown before login	banner motd

8.2 Configuration

The same block was applied to EDGE-R1, CORE-SW, SW-ADMIN, SW-STUDIO and SW-PROD:

```
ip domain-name platinumsound.local
username netadmin privilege 15 secret <redacted>
enable secret <redacted>
service password-encryption
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
access-list 10 remark SSH allowed from VLAN 10 only
access-list 10 permit 192.168.52.0 0.0.0.31
line vty 0 15
 transport input ssh
 login local
 access-class 10 in
 exec-timeout 5 0
line console 0
 login local
 exec-timeout 5 0
banner motd # Authorised access only - Platinum Sound Studio #
```

A hostname and domain name must exist before the RSA keys can be generated, because the key is named after them for example `CORE-SW.platinumsound.local`. On EDGE-R1, SSH uses access list 10 while NAT uses access list 1, so the two policies cannot interfere.

8.3 Scope beyond the Milestone 1 design

The Milestone 1 design named only the router and core switch for SSH. SSH was also applied to the three access switches because the requirements state that no infrastructure device may run a cleartext protocol. An access switch left on Telnet would have been the weak point.

8.4 Verification

Test	Result
`show ip ssh` on CORE-SW	SSH Enabled – version 2.0, timeout 60 s, 3 retries
SSH from ADMIN-PC1 to CORE-SW, EDGE-R1, SW-STUDIO	Login succeeded at each device
`show ssh` during a session	aes128-cbc encryption, hmac-sha1 integrity, user netadmin
Telnet from ADMIN-PC1 to CORE-SW	Connection closed with no login prompt
SSH from STUDIO-PC1 to CORE-SW	"Connection refused by remote host", no password prompt

The `show ssh` session table labels the session "1.99", which is how Packet Tracer reports the negotiated protocol; `show ip ssh` confirms the server itself runs version 2.0 only.

9. Testing and verification

All 15 tests passed. Each phase was tested before the next was built, and every security control was tested from both sides. Once where it should allow traffic and once where it should block it.

9.1 Method

- **Connectivity -** `ping` between VLANs and to the internet server, TTL values confirm the number of routed hops.
- **Services -** DHCP leases checked on every PC, DNS and HTTP checked by loading www.example.com; FTP login to the file server.
- **Device state -** `show ip interface brief`, `show vlan brief`, `show interfaces trunk`, `show ip nat translations`.
- **Security -** before-and-after tests for the guest ACL with `show access-lists` hit counters; SSH and Telnet attempts from allowed and blocked VLANs.

9.2 Results

#	Test	From → To	Expected	Result
1	VLANs and trunks configured	Access switches	Accepted, trunks up	Pass
2	Gateways up	CORE-SW	Vlan10–50 up/up	Pass
3	DHCP in all user VLANs	6 PCs + contractor	Correct pool and gateway	Pass
4	Inter-VLAN routing	ADMIN-PC1 → STUDIO-PC1	Reply, TTL 127 (1 hop)	Pass
5	Internet reachability	ADMIN-PC1 → 198.51.100.10	Reply, TTL 125 (3 hops)	Pass
6	DNS and web	ADMIN-PC1 → www.example.com	Page loads	Pass
7	NAT overload	EDGE-R1	192.168.52.10 → 203.0.113.2	Pass
8	File server	PROD-PC2 → FILE-SRV	Ping and FTP login	Pass
9	Guest baseline, before ACL	CONTRACTOR-1 → FILE-SRV	Reply (gap exists)	Pass

#	Test	From → To	Expected	Result
10	Guest isolation, after ACL	CONTRACTOR-1 → internal	Dropped, 8 ACL matches	Pass
11	Guest internet, after ACL	CONTRACTOR-1 → internet	DNS, HTTP, ping allowed	Pass
12	SSHv2 enabled	CORE-SW	Version 2.0	Pass
13	SSH from management VLAN	ADMIN-PC1 → .1, .129, .3	Encrypted login	Pass
14	Telnet blocked	ADMIN-PC1 → .1	No login prompt	Pass
15	SSH from other VLANs blocked	STUDIO-PC1 → .1	Connection refused	Pass

Screenshots for every test are in the repository's `/testing` folder and in the Milestone 2 Implementation Review.

9.3 Troubleshooting

Problem	Cause	Fix
First pings to a new destination timed out	ARP resolving MAC addresses at each hop	Normal behaviour; later pings replied
Router CLI froze for about a minute	A stray word was read as a hostname and IOS tried to look it up in DNS	Power-cycled the unconfigured router; `no ip domain-lookup` added to every device
SW-STUDIO saved an empty configuration	The Config tab's Save ran before the commands were entered	Applied the configuration, then saved again
Last command of a paste did not run	No line break after the final command	Pressed Enter after every paste and checked the prompt
Contractor DHCP line showed 0 ACL matches	The phone leased its address before the ACL was applied	Expected; renewing the lease would match line 10

10. Reflection

Building and testing in small phases was the decision that mattered most. Every fault I hit was in the step I had just done so it took minutes not hours to find.

10.1 What went well

- **Design held up in the build -** The Milestone 1 addressing plan, VLANs and gateways were implemented without a single change which showed the planning was sound.
- **Testing both sides of each control -** Recording the contractor reaching the file server before the ACL made the after result meaningful. Testing SSH from both an allowed and a blocked VLAN proved the access-class works, not just that SSH runs.
- **Evidence as I went -** Taking screenshots and committing to GitHub after each phase meant the portfolio grew with the work instead of being rebuilt at the end.

10.2 Challenges

- **Packet Tracer limits -** The `time-range` command is not supported so the after-hours part of CR-14 could only be documented, not demonstrated. I learned to separate what the simulator can show from what real IOS would do.
- **Small CLI mistakes with big effects -** A mistyped word froze the router while IOS tried to resolve it as a hostname. Adding `no ip domain-lookup` to every device early on would have avoided it.
- **ACL order -** The guest ACL only works because DHCP is permitted before the internal deny and the internet permits come after it. Getting the order wrong would have either blocked contractors completely or let them into the studio network.

10.3 What I would do differently

- Put `no ip domain-lookup` and a hostname into a standard first block for every device.
- Use a dedicated management VLAN instead of sharing VLAN 10 with admin users, so a compromised admin PC could not reach the device management lines.
- Add a second link between the core and each access switch or a second core switch so one cable or switch failure does not cut off a whole department.
- Use 2048-bit RSA keys on real hardware, 1024 bits meets the SSHv2 minimum but is no longer considered strong.

11. Conclusion

The network meets all five functional requirements and passes all 15 tests so it is ready for client acceptance. Staff areas are separated, session files are central and reachable, the internet works through NAT, contractors are confined to internet-only Wi-Fi and every infrastructure device is managed over encrypted SSH from the management VLAN only.

Two items are noted for the client:

- The CR-14 after-hours time limit is designed and documented but needs real Cisco hardware to run, because Packet Tracer cannot simulate it.
- The improvements in section 10.3 a dedicated management VLAN, redundant uplinks and stronger SSH keys are recommended before the network goes into production.

12. Appendices

A. Deliverables

Deliverable	Location
Packet Tracer project	`52827097_SEDZE_CMPG325-2026-121.pkt` in the repository root
Running configurations	`/configs`: EDGE-R1, CORE-SW, SW-ADMIN, SW-STUDIO, SW-PROD
Testing screenshots	`/testing`, numbered to match the Implementation Review
Milestone 1 Client Design Review	`/docs`
Milestone 2 Implementation Review	`/docs`
Video demonstration (15 - 20 min, inset)	Link in the repository README
GitHub repository	https://github.com/wantedProgrammer/Computer-Networks

B. Device credentials

Credentials are not published in this report or the repository. The local SSH account is `netadmin`; passwords are stored as hashes on every device and are available to the client separately.

C. References

- Cisco Networking Academy, *Cisco Packet Tracer*
- RFC 5737, *IPv4 Address Blocks Reserved for Documentation*
- RFC 4253, *The Secure Shell (SSH) Transport Layer Protocol* (SSHv2).